Improved Codes and Decoders for HQC

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Abstract. The Hamming Quasi-Cyclic (HQC) cryptosystem, recently selected by NIST for standardization, uses a concatenation of Reed–Muller (RM) and Reed–Solomon (RS) codes to ensure correct decryption. This work optimizes the underlying code and decoder with the goal of reducing ciphertext and public-key sizes. First, we propose a two-level generalized concatenated code (GCC), exploiting the observation that the inner RM code contains a repetition subcode. The repetition layer incurs a smaller inner failure probability and, therefore, admits a higher code rate than the RS code. Second, we develop a reliability-based decoding framework for the outer code. The inner decoder provides a reliability measure whose distribution, jointly with the decoding outcome, is bounded analytically. These bounds allow deriving conservative DFR guarantees for two erasure-assignment strategies (threshold-based and partition-based). Combining both contributions, we observe that public-key and ciphertext sizes can be reduced by up to 4.34 %.

1 Introduction

Hamming Quasi-Cyclic (HQC) [11] is a code-based key-encapsulation mechanism whose security relies on the hardness of decoding quasi-cyclic codes in the Hamming metric. Following several rounds of evaluation, HQC was selected by NIST as the fifth post-quantum algorithm [15]. To encrypt a message $\mathbf{m} \in \mathbb{F}_2^\lambda$, it is first encoded via an efficient error-correcting code $\mathcal{C}_{\text{pub}}$ before applying a random-looking mask. During decryption, the mask is removed using the secret key, but a residual error $\mathbf{e}$ of moderate Hamming weight remains. To achieve IND-CCA2 security and thwart attacks based on decryption failures [12,13], the decoder of $\mathcal{C}_{\text{pub}}$ must correct $\mathbf{e}$ with probability $1 - 2^{-\lambda}$, where λ denotes the bit security level (e.g., $\lambda = 128$ for NIST security level 1).

The Decryption Failure Rate (DFR) analysis of [11] models the HQC error $\mathbf{e}$ as a vector sampled from a Bernoulli distribution with an error probability p, which is inversely proportional to the code length n. The public-key size is n bits and the ciphertext size is $2n$ bits, so reducing n directly reduces both. Under this model, an analytical DFR bound for a hard-decision decoder of a concatenated code was derived in [11]. Initially, HQC used the concatenation of an inner repetition and an outer BCH code [3]. In the current version, this was replaced by an inner (duplicated) Reed–Muller and outer Reed–Solomon code, resulting in shorter codes and better runtime.

1.1 Contributions

We propose two complementary techniques to reduce HQC's public-key and ciphertext sizes, achieving a reduction of up to 4.34 %.

Generalized concatenated codes. We observe that the inner RM code currently used in HQC contains the repetition code as a subcode, forming a natural partition chain (Section 3). This motivates a 2-level generalized concatenated code (GCC) [5, 19], which can be interpreted as combining the round-one (repetition-BCH) and round-two (RM-RS) constructions. The repetition layer has twice the minimum distance of the RM layer and therefore incurs a much smaller inner failure probability, admitting a higher-rate outer code. The additional information bits carried at this level allow shortening the overall code while maintaining the required DFR and message length. Section 3.1 provides a general overview of GCCs and their DFR analysis [16]; Section 3.2 instantiates the framework for HQC with a concrete 2-level construction.

Reliability-based outer decoding. The current HQC specification [11] uses a hard-decision outer decoder, which does not exploit the confidence information available from the inner decoding. We develop a framework for errors-and-erasures outer decoding, in which positions with low inner-decoding confidence are declared as erasures (Section 4). Since an errors-and-erasures decoder can correct w errors and ϵ erasures whenever $2w + \epsilon < d$, converting suspected errors into erasures effectively doubles their correction efficiency.

The framework is based on a reliability measure derived from the inner code's geometry. We bound its distribution jointly with the decoding outcome (Section 4.1), and use these bounds to derive rigorous DFR guarantees for two erasure-assignment strategies: a *threshold-based* scheme, where all symbols with reliability below a fixed threshold are erased (Section 4.2), and a *partition-based* scheme, where a fixed-size subset of the least reliable symbols is erased (Section 4.3). The DFR analysis uses stochastic dominance and order-statistic techniques, respectively.

Reproducibility. Scripts for reproducing computations and simulations are available at `https://gitlab.com/bharath.purtipli/hqc_gcc`.

1.2 Related Work

Aguilar-Melchor et al. [2] introduced the current RM-RS concatenated coding scheme for HQC, replacing the earlier BCH-repetition construction. They provide a detailed DFR analysis of the duplicated first-order RM code under hard-decision RS decoding. Our work builds on their construction by extending it to a GCC and by adding reliability-based outer decoding.

HARE [17] is a recently proposed adaptation of the HQC standard that achieves significant reductions in ciphertext and public-key sizes through ciphertext compression, unbalanced encryption weights, and an errors-and-erasures RS decoder. Their errors-and-erasures decoder uses the same reliability measure and threshold-based erasure assignment as Section 4.2 of this work, with a DFR analysis specific to that setting. In contrast, we develop a general analytical framework for the reliability measure (Section 4.1) that supports multiple erasure strategies (Sections 4.2 and 4.3) and additionally propose a GCC structure (Section 3).

GMD decoding of HQC's RS outer code was proposed in [8]. While their approach is well-motivated, the DFR bounds rely on extrapolation from simulation data, which may not yield valid guarantees for DFRs of $2^{-\lambda}$. In contrast, our DFR analysis is fully analytical and yields provable upper bounds.

2 Preliminaries

2.1 Notation

We write $[n] \triangleq \{1, \ldots, n\}$. Integers and field elements are denoted by lower-case Roman letters, vectors by bold lower-case, and matrices by bold upper-case. Denote by $\mathbb{F}_q$ the finite field with q elements. For $\mathbf{u}, \mathbf{v} \in \mathbb{F}_q^n$, the Hamming distance is $\mathrm{d_H}(\mathbf{u}, \mathbf{v}) \triangleq |\{i \in [n] : u_i \neq v_i\}|$ and the Hamming weight is $|\mathbf{v}|_{\mathrm{H}} \triangleq \mathrm{d_H}(\mathbf{v}, \mathbf{0})$. Let $\mathcal{R}_n \triangleq \mathbb{F}_2[x]/(x^n - 1)$; for $\mathbf{h}, \mathbf{y} \in \mathcal{R}_n$, $\mathbf{h} \cdot \mathbf{y}$ denotes the product in $\mathcal{R}_n$.

2.2 Codes and Decoders

A *linear code* $\mathcal{C}$ with parameters $[n, k, d]_q$ is a k-dimensional subspace of $\mathbb{F}_q^n$ with minimum Hamming distance d. For given $\mathcal{C}$, we also write $k(\mathcal{C})$ to refer to its dimension and $d(\mathcal{C})$ to refer to its minimum distance. An *encoder* $\mathsf{enc}(\cdot)$ for $\mathcal{C}$ is a bijective linear map $\mathbb{F}_q^k \to \mathcal{C}$, characterized by a generator matrix $\mathbf{G} \in \mathbb{F}_q^{k \times n}$ with $\mathsf{enc}(\mathbf{m}) = \mathbf{mG}$.

Decoders. A *decoder* $\mathsf{dec}(\cdot)$ maps a received vector $\mathbf{v} \in \mathbb{F}_q^n$ to a codeword $\hat{\mathbf{c}} \in \mathcal{C}$. A *maximum-likelihood (ML)* decoder returns $\hat{\mathbf{c}} = \arg\min_{\mathbf{c} \in \mathcal{C}} \mathrm{d_H}(\mathbf{v}, \mathbf{c})$, with ties broken by a fixed deterministic rule. A *bounded-minimum-distance (BMD)* decoder successfully decodes any error pattern with $|\mathbf{e}|_{\mathrm{H}} \leq \lfloor (d-1)/2 \rfloor$.

An *errors-and-erasures decoder* accepts input $\mathbf{v} \in (\mathbb{F}_q \cup \{\Delta\})^n$, where Δ denotes an erasure (unknown symbol value with known position). A BMD errors-and-erasures decoder succeeds whenever $2\tau + \epsilon < d$, where ϵ is the number of erasures and τ the number of errors among the non-erased positions.

Reed–Solomon (RS) codes. For any prime power $q \geq n$, there exists an RS code with parameters $[n, k, n-k+1]_q$. RS codes are maximum-distance separable (MDS) and admit efficient BMD errors-and-erasures decoding [7].

Reed–Muller (RM) codes. The r-th order binary Reed–Muller code $\mathrm{RM}(r, m)$ has length 2^m, dimension $\sum_{i=0}^{r} \binom{m}{i}$, and minimum distance 2^{m-r}. For $r = 0$, this is the binary repetition code $\mathrm{Rep} \triangleq \{(0, \ldots, 0), (1, \ldots, 1)\}$ of length 2^m. For $r = 1$, ML decoding can be performed efficiently via the Fast Walsh–Hadamard Transform (FWHT).

Duplicated RM codes. Following [2, 11], we use the tensor product $\mathrm{Rep} \otimes \mathrm{RM}(r, m)$, which repeats every bit of an $\mathrm{RM}(r, m)$ codeword δ times. For duplication factor δ, the resulting code has parameters $[\delta \cdot 2^m,\ \sum_{i=0}^{r} \binom{m}{i},\ \delta \cdot 2^{m-r}]_2$.

ML decoding first accumulates the δ copies of each bit, then applies the FWHT to decode the RM component. A bound on the ML failure probability is provided in [2,11].

2.3 HQC in a Nutshell

We provide a brief overview of the HQC public-key encryption scheme, focusing on the error-correction mechanism that forms the center of this work. For cryptographic details, refer to [11].

HQC uses a linear $[n_{\text{out}}n_{\text{in}}, k = \lambda, d_{\text{pub}}]_2$ code $\mathcal{C}_{\text{pub}}$ with a generator matrix $\mathbf{G}_{\text{pub}}$. The secret key is a random pair of moderately sparse vectors $\mathbf{x}, \mathbf{y} \in \mathcal{R}_n$ each of Hamming weight $w_{\mathtt{sk}}$. The corresponding public key is the pair of vectors $\mathbf{h}$ and $\mathbf{s} = \mathbf{x} + \mathbf{h} \cdot \mathbf{y}$.

Encryption. To encrypt a message $\mathbf{m} \in \mathbb{F}_2^{\lambda}$, one creates a pair of vectors $\mathbf{u} = \mathbf{r}_1 + \mathbf{h} \cdot \mathbf{r}_2$ and $\mathbf{v} = \mathbf{m}\mathbf{G}_{\text{pub}} + \mathbf{s} \cdot \mathbf{r}_2 + \mathbf{r}_3$ where $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3 \in \mathcal{R}_n$ are random vectors of weight $w_{\mathtt{ct}}$.

Decryption. Decryption applies an efficient decoder of $\mathcal{C}_{\text{pub}}$ on $\mathbf{v} + \mathbf{u} \cdot \mathbf{y} = \mathbf{m}\mathbf{G}_{\text{pub}} + \mathbf{e}$ where the residual error is $\mathbf{e} = \mathbf{x} \cdot \mathbf{r}_2 + \mathbf{y} \cdot \mathbf{r}_1 + \mathbf{r}_3$. The message $\mathbf{m}$ is recovered by applying the decoder of $\mathcal{C}_{\text{pub}}$. Since $\mathbf{e}$ is random, decryption may fail; the code parameters are chosen so that this *decryption failure rate* (DFR) is at most $2^{-\lambda}$.

Error model. The error $\mathbf{e}$ can be modeled as having i.i.d. Bern(p) entries, i.e., errors occur independently with probability p. As shown in [11], p depends on the system parameters via

$$p = \Pr\left[\mathbf{e}_i = 1\right] = 2(1-\tilde{p})\tilde{p}\left(1 - \tfrac{w_{\mathtt{ct}}}{n}\right) + \left((1-\tilde{p})^2 + \tilde{p}^2\right)\frac{w_{\mathtt{ct}}}{n},$$

where

$$\tilde{p} = \binom{n}{w_{\mathtt{ct}}}^{-1} \cdot \sum_{\substack{1 \le \ell \le \min\{w_{\mathtt{sk}}, w_{\mathtt{ct}}\} \\ \ell \text{ odd}}} \binom{w_{\mathtt{sk}}}{\ell}\binom{n - w_{\mathtt{sk}}}{w_{\mathtt{ct}} - \ell}.$$

The crossover probability p is inversely proportional to n: longer codes operate at a lower noise level. Alternative error models that account for dependencies in the error $\mathbf{e}$ have been studied in [4,6].

3 More Efficient Error-Correcting Code

This section provides an overview on generalized concatenated codes (Section 3.1) and describes the code construction tailored to HQC (Section 3.2).

3.1 An Overview on Generalized Concatenated Codes

Zinoviev [19] and Blokh and Zyablov [5] provided a generalization of concatenated codes to several levels: whereas standard concatenation pairs a single outer code with a single inner code, a GCC associates a separate outer code with each level of an inner code partition chain. For a thorough introduction to generalized code concatenation, also known as multi-level concatenation, see [9] or [7], and references therein.

To construct a binary s-level GCC, select a nested sequence of inner codes

$$\{\mathbf{0}\} = \mathcal{B}_{s+1} \subseteq \mathcal{B}_s \subseteq \cdots \subseteq \mathcal{B}_1 \subseteq \mathbb{F}_2^{n_{\mathrm{in}}}.$$

Note that the indexing convention has $\mathcal{B}_1$ as the *largest* (and weakest) inner code and $\mathcal{B}_s$ as the smallest (and strongest). Correspondingly, the inner minimum distance $d(\mathcal{B}_j)$ is non-decreasing in j. By abuse of notation, $\mathcal{B}_j/\mathcal{B}_{j+1} \subset \mathbb{F}_q^{n_{\mathrm{in}}}$ denotes the (not necessarily unique) linear code of dimension $k(\mathcal{B}_j) - k(\mathcal{B}_{j+1})$ such that

$$\mathcal{B}_j = \{\mathbf{c}_1 + \mathbf{c}_2 \mid \mathbf{c}_1 \in \mathcal{B}_{j+1}, \mathbf{c}_2 \in \mathcal{B}_j/\mathcal{B}_{j+1}\}.$$

Before introducing GCCs using this notation, we illustrate it by partitioning the inner code used in HQC's code concatenation [11].

Example 1. Let $\mathcal{B}_1$ be a duplicated first-order RM code of dimension k_{RM} as used in [11]. $\mathcal{B}_1$ is generated by $\langle \mathbf{g}_1, \ldots, \mathbf{g}_{k_{\mathrm{RM}}} \rangle$ with $\mathbf{g}_1 = (1, \ldots, 1)$. Hence, $\mathcal{B}_1$ contains the length-n_{in} repetition code as a subcode $\mathcal{B}_2$. Then, we may select $\mathcal{B}_1/\mathcal{B}_2 = \langle \mathbf{g}_2, \ldots, \mathbf{g}_{k_{\mathrm{RM}}} \rangle$.

Definition 1. *Let a nested sequence of inner codes $\{\mathbf{0}\} = \mathcal{B}_{s+1} \subseteq \mathcal{B}_s \subseteq \cdots \subseteq \mathcal{B}_1 \subseteq \mathbb{F}_2^{n_{\mathrm{in}}}$ be given. Let outer codes $\mathcal{A}_j \subseteq \mathbb{F}_{2^{m_j}}^{n_{\mathrm{out}}}$ with $m_j = k(\mathcal{B}_j) - k(\mathcal{B}_{j+1})$ be given for all $j \in [s]$. The* generalized concatenated code GCC *is constructed as*

$$\mathrm{GCC} = \left\{ \Big(\sum_{j\in[s]} (\mathcal{B}_j/\mathcal{B}_{j+1}).\mathsf{enc}(a_{j,\ell}) \Big)_{\ell\in[n_{\mathrm{out}}]} \;\middle|\; \mathbf{a}_j \in \mathcal{A}_j \;\forall j \in [s] \right\}.$$

Then, GCC *has length $n_{\mathrm{out}} \cdot n_{\mathrm{in}}$ and dimension $\sum_{j\in[s]} k(\mathcal{A}_j)$.*

The following example highlights that Definition 1 specializes to a standard concatenated code (tensor product) for $s = 1$, as currently deployed in HQC [11].

Example 2. To construct the concatenated code proposed in [11] for NIST 1, we set $s = 1$. As $\mathcal{B}_1$, pick a first-order duplicated RM code with parameters $[384, 8, 192]_2$. As $\mathcal{A}_1$, a RS code with parameters $[46, 16, 31]_{2^8}$ is used, with alphabet $\mathbb{F}_{2^8}$ matching the inner code dimension. To construct the error-correcting codes at NIST 3 and NIST 5, the length of the inner RM code and the dimension of the outer RS code are increased.

The minimum distance of a GCC is lower-bounded as

$$d(\mathrm{GCC}) \geq \min_{j\in[s]} d(\mathcal{A}_j) \cdot d(\mathcal{B}_j), \tag{1}$$

As the inner minimum distance $d(\mathcal{B}_j)$ is increasing in j, it is natural to have $d(\mathcal{A}_j)$ decrease with j (allowing higher-dimensional outer codes).

In the following, we recall the standard decoding algorithm for GCCs. Instantiated with suitable decoders for the component codes, it is guaranteed to correct all errors up to half the bound in Equation (1).

GCC decoder. Let $\mathbf{r} = (\mathbf{r}_1, \dots, \mathbf{r}_{n_{\text{out}}}) = \mathbf{c}+\mathbf{e}$ be given with codeword $\mathbf{c} \in \text{GCC}$ and error $\mathbf{e}$ sampled i.i.d. from $\text{Bern}(p)$ (cf. Section 2.3). The decoder proceeds level by level in a multistage fashion. For $j = 1, \dots s$:

1. **Inner decoding at level j.** A decoder of $\mathcal{B}_j$ is applied to each $\mathbf{r}_i$ to obtain an estimated inner codeword $\hat{\mathbf{c}}_i \in \mathcal{B}_j$. Optionally (see Section 4), a reliability measure α_i indicating the confidence of the inner decoding may be derived from the geometry of $\mathbf{r}_i$ and $\mathcal{B}_j$.
2. **Outer decoding at level j.** Each $\hat{\mathbf{c}}_i \in \mathcal{B}_j$ corresponds to an estimated symbol of the outer code $\mathcal{A}_j$. To correct errors in this sequence of estimated symbols, a decoder of $\mathcal{A}_j$ is applied, which may utilize the reliabilities α_i. This yields an estimated outer codeword $\hat{\mathbf{a}}_j$ (or possibly a decoding failure).
3. **Cancellation.** The estimated codeword $\hat{\mathbf{a}}_j$ is re-encoded as

$$\hat{\mathbf{c}}_j = \big((\mathcal{B}_j/\mathcal{B}_{j+1}).\mathsf{enc}(a_{j,\ell})\big)_{\ell\in[n_{\text{out}}]}\,.$$

   Before continuing with level $j+1$, $\hat{\mathbf{c}}_j$ is canceled from $\mathbf{r}_j$ via $\mathbf{r}_j - \hat{\mathbf{c}}_j$.

DFR analysis The described decoder fails if and only if the outer decoding fails on at least one level. Denote by $E_j = \{\hat{\mathbf{a}}_j \neq \mathbf{a}_j\}$ the decoding failure event at level j. Then, using the union bound [16], an upper bound for the DFR is obtained as

$$\text{DFR} = \Pr\Big[\bigcup\nolimits_{j\in[s]} E_j\Big] \leq \sum_{j\in[s]} \Pr\big[E_j\big]. \tag{2}$$

This allows analyzing the failure probability at each level separately. Parameters are chosen such that $\sum_{j\in[s]} \Pr\big[E_j\big] \leq 2^{-\lambda}$. This simple possibility of analyzing the DFR makes GCCs a suitable candidate for error-correction in HQC. In the following section, we propose a concrete instantiation of Definition 1 with two levels, which builds on the construction in Example 2.

3.2 A Two-Level GCC for HQC

As illustrated in Example 2, the error correcting code of HQC is currently instantiated as a concatenation of an inner RM and an outer RS code. Previously, the concatenation of an inner repetition code with an outer binary BCH code was used [3], which was however less efficient.

In Example 1, we have observed that the repetition code and the duplicated first-order RM code form a partition chain. This motivates the 2-level GCC construction for $\mathcal{C}_{\text{pub}}$ described in Table 1. As in [11], we pair the RM code with an outer RS code at level 1. The repetition code at level 2 requires a binary outer code; as in [3], we use a BCH code to achieve efficient hard-decision or errors-and-erasure decoding (cf. Section 4). Both the repetition and the RM code can be efficiently maximum-likelihood decoded.

Table 1: Code structure of the proposed 2-level GCC.

Level j	Outer Code $\mathcal{A}_j$		Inner Code $\mathcal{B}_j$	
1	RS	$[n_{\text{out}}, k_{\text{RS}}, d_{\text{RS}}]_{2^m}$	RM	$[n_{\text{in}}, k_{\text{RM}}, \frac{n_{\text{in}}}{2}]_2$
2	$\mathcal{C}$	$[n_{\text{out}}, k_{\mathcal{C}}, d_{\mathcal{C}}]_2$	Rep	$[n_{\text{in}}, 1, n_{\text{in}}]_2$

Performance improvement. The key advantage of the 2-level construction is that the repetition code (level 2) has minimum distance n_{in}, which is twice the RM code's minimum distance. Consequently, the inner decoding error probability at level 2 is significantly smaller than at level 1. The corresponding binary outer code $\mathcal{C}$ therefore requires a smaller minimum distance, and can operate at a higher rate. Since the overall message length is fixed to λ, this improvement translates to a smaller RS dimension or, equivalently, a shorter overall code length. This effect reduces the size of the public key and the ciphertext.

The achievable improvement depends on the outer decoder; the following example illustrates the interplay between the GCC structure and the decoder for NIST 1 parameters.

Example 3. The current HQC specification [11] uses a single-level concatenation with an outer $[46, 16, 31]_{2^8}$ RS code and an inner $[384, 8, 192]_2$ RM code, giving $n = 17669$. At $p = 0.340$, a DFR of $2^{-132.9}$ is achieved using inner ML and outer hard-decision decoding.

Using the improved errors-and-erasures decoder of Section 4, we can reduce the outer length to $n_{\text{out}} = 44$ (and correspondingly $n = 16901$, reducing the public-key size by 4.34 %). At this shorter length, the DFR analysis requires an RS minimum distance of $d_{\text{RS}} = 30$, leaving $k_{\text{RS}} = n_{\text{out}} - d_{\text{RS}} + 1 = 15$ information symbols. For a single concatenation level, this yields $15 \cdot 8 = 120$ information bits, which is insufficient for $\lambda = 128$.

The 2-level GCC resolves this shortfall. At level $j = 2$, the repetition code has minimum distance $n_{\text{in}} = 384$, twice that of the RM code. This results in a much smaller inner failure probability, so that an outer binary BCH code with minimum distance $d_{\mathcal{C}} = 8$ suffices. A shortened $[n_{\text{out}}, 25, 8]_2$ BCH code contributes $k_{\mathcal{C}} = 25$ information bits, for a total of $15 \cdot 7 + 25 = 130 \geq 128 = \lambda$.

Refer to Section 5 for further parameter suggestions, also relying on the improved decoding presented in Section 4, which is fully compatible with the 2-level construction.

On adding a third level via RM(2,m). Higher order RM codes are supercodes of the duplicated first-order RM code. It is therefore a natural question whether it makes sense to extend the code presented in Table 1 to $s = 3$ levels via the partition chain

$$\text{Rep} \subset \text{RM}(1, m) \subset \text{RM}(2, m),$$

where the minimum distance of $\text{RM}(2, m)$ is half the minimum distance of $\text{RM}(1, m)$. To estimate the failure probability of $\text{RM}(2, m)$ under maximum-likelihood decoding, we use the union bound

$$\bar{P}_{\text{in}} \triangleq \sum_{w=1}^{n_{\text{in}}} A_w \sum_{i=\lceil w/2 \rceil}^{w} \binom{w}{i} p^i (1-p)^{w-i},$$

where $A_0, \dots A_{n_{\text{in}}}$ denotes the weight distribution of the code, which is known for $\text{RM}(2,m)$ [14]. For parameters suggested in [11], we find $\bar{P}_{\text{in}} > 1$. While this does not directly imply that the true error probability equals 1 (the union bound may be loose), it indicates that the inner code operates far beyond its error-correction capability at this noise level. Consequently, the error-correction capability of the corresponding outer code would need to be enormous, limiting the number of information bits that can be allocated to this level. Due to this limitation, and the computational cost of actually decoding $\text{RM}(2,m)$, we restrict ourselves to the 2-level construction shown in Table 1.

4 Error-Erasure Decoder for HQC

This section analyzes errors-and-erasures decoding of the outer code. The reliability measure that we derive from the inner decoding is inspired by [1], and we conservatively bound its distribution in Section 4.1. Section 4.2 considers an erasure assignment scheme in which all symbols with the reliabilities below a predetermined threshold are erased. In Section 4.3, we consider the approach of erasing the ϵ least reliable symbols. All results are stated for a single concatenation level and assuming a generic inner code $\mathcal{B}$; the extension to multiple levels via Equation (2) is immediate. Without loss of generality, we assume $\mathbf{c}_i = \mathbf{0}$ (so $\mathbf{r}_i = \mathbf{e}_i$) throughout the section.

4.1 Analysis of Proposed Reliability Measure

In addition to the estimated symbol, the inner decoding inherently provides reliability information. For example, when $\hat{\mathbf{c}}_i = \mathbf{r}_i$, the inner decoding is correct with high probability. On the other hand, if there is another codeword $\hat{\mathbf{c}}_i'$ with $\text{d}_{\text{H}}(\hat{\mathbf{c}}_i, \mathbf{r}_i) = \text{d}_{\text{H}}(\hat{\mathbf{c}}_i', \mathbf{r}_i)$, the inner decoding is correct with probability at most $\frac{1}{2}$. More generally, the posterior probability of the decoded codeword given the received word equals

$$\Pr[\hat{\mathbf{c}}_i \mid \mathbf{r}_i] = \frac{\Pr[\mathbf{r}_i \mid \hat{\mathbf{c}}_i]}{\sum_{\mathbf{c}\in\mathcal{C}} \Pr[\mathbf{r}_i \mid \mathbf{c}]} = \frac{(\frac{p}{1-p})^{\text{d}_{\text{H}}(\hat{\mathbf{c}}_i, \mathbf{r}_i)}}{\sum_{\mathbf{c}\in\mathcal{C}} (\frac{p}{1-p})^{\text{d}_{\text{H}}(\mathbf{c}, \mathbf{r}_i)}}$$

and constitutes the optimal reliability measure.

Example 4. For a binary repetition code of length n_{in}, we have

$$\Pr[\hat{\mathbf{c}}_i \mid \mathbf{r}_i] = \frac{(\frac{p}{1-p})^{\text{d}_{\text{H}}(\hat{\mathbf{c}}_i, \mathbf{r}_i)}}{\sum_{\mathbf{c}\in\mathcal{C}} (\frac{p}{1-p})^{\text{d}_{\text{H}}(\mathbf{c}, \mathbf{r}_i)}} = \left(1 + (\frac{p}{1-p})^{d_i' - d_i}\right)^{-1},$$

where $d_i = \mathrm{d_H}(\hat{\mathbf{c}}_i, \mathbf{r}_i)$ denotes the distance to the estimated codeword $\hat{\mathbf{c}}_i$ and $d'_i = \mathrm{d_H}(\hat{\mathbf{c}}'_i, \mathbf{r}_i)$ the distance to the second-best one. As $(\frac{p}{1-p}) < 1$, $\Pr[\hat{\mathbf{c}}_i \,|\, \mathbf{r}_i]$ is strictly increasing in $\alpha_i = d'_i - d_i$, which is therefore a suitable choice of reliability measure, i.e., it captures all information contained in $\Pr[\hat{\mathbf{c}}_i \,|\, \mathbf{r}_i]$. Moreover, $d'_i = n_{\mathrm{in}} - d_i$.

Example 4 illustrates that precisely characterizing the distribution of the reliability information (also conditioning on erroneous or successful decoding) is straightforward for binary repetition codes, since there are only two codewords. For general inner codes, however, such a characterization requires accounting for the complete code geometry, which quickly becomes infeasible as $|\mathcal{B}|$ grows. To overcome this issue, observe that typically the best estimate $\hat{\mathbf{c}}_i$ and the second-best $\hat{\mathbf{c}}'_i$ are closer to $\mathbf{r}_i$ than any other codeword by a significant margin. Therefore, we may approximate $\sum_{\mathbf{c}\in\mathcal{B}} \Pr[\mathbf{r}_i \,|\, \mathbf{c}] \approx \Pr[\mathbf{r}_i \,|\, \hat{\mathbf{c}}_i] + \Pr[\mathbf{r}_i \,|\, \hat{\mathbf{c}}'_i]$, which yields

$$\Pr[\hat{\mathbf{c}}_i \,|\, \mathbf{r}_i] \approx \frac{1}{1 + (\frac{p}{1-p})^{\alpha_i}} \text{ with } \alpha_i \triangleq d'_i - d_i = \mathrm{d_H}(\hat{\mathbf{c}}'_i, \mathbf{r}_i) - \mathrm{d_H}(\hat{\mathbf{c}}_i, \mathbf{r}_i),$$

which is monotonically increasing in α_i, justifying our choice of α_i as the reliability measure.

Remark 1. This approximation is exact for repetition codes and tight whenever the two nearest codewords dominate the likelihood sum; the formal analysis below via $\tilde{\alpha}_i$ and union bounds does not rely on this approximation for correctness.

Analysis of distribution. Looking ahead, the DFR analyses in Sections 4.2 and 4.3 require the distribution of α_i jointly with the decoding outcome, i.e., the joint probabilities $\Pr\big[\alpha_i \geq x, \hat{\mathbf{c}}_i = \mathbf{c}_i\big]$ and $\Pr\big[\alpha_i \geq x, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big]$. Since the joint distribution of α_i and the decoding outcome depends on the full distance profile of $\mathcal{B}$, we define an alternative quantity $\tilde{\alpha}_i$ that admits a tractable bound.

Lemma 1. *Define* $\hat{\mathbf{c}}^\star_i \triangleq \arg\min_{\mathbf{v}\in\mathcal{B}\setminus\{\mathbf{c}_i\}} \mathrm{d_H}(\mathbf{v}, \mathbf{r}_i)$ *as the closest codeword in* $\mathcal{B} \setminus \{\mathbf{c}_i\}$, $w_i = |\mathbf{e}_i|_{\mathrm{H}}$, *and* $w^\star_i = \mathrm{d_H}(\hat{\mathbf{c}}^\star_i, \mathbf{r}_i)$. *Then*

$$\tilde{\alpha}_i \triangleq |w^\star_i - w_i| \begin{cases} = \alpha_i & \textit{if } \hat{\mathbf{c}}_i = \mathbf{c}_i \\ \geq \alpha_i & \textit{otherwise.} \end{cases}$$

Proof. For $\hat{\mathbf{c}}_i = \mathbf{c}_i$, we have $w_i = \mathrm{d_H}(\mathbf{r}_i, \hat{\mathbf{c}}_i) = d_i$ and $w^\star_i = \mathrm{d_H}(\mathbf{r}_i, \hat{\mathbf{c}}'_i) = d'_i$. For $\hat{\mathbf{c}}_i \neq \mathbf{c}_i$, $\hat{\mathbf{c}}^\star_i = \hat{\mathbf{c}}_i$ (since $\hat{\mathbf{c}}_i \in \mathcal{B} \setminus \{\mathbf{c}_i\}$ minimizes $\mathrm{d_H}(\mathbf{v}, \mathbf{r}_i)$ over all $\mathbf{v} \in \mathcal{B}$, hence also over $\mathcal{B} \setminus \{\mathbf{c}_i\}$), so $w^\star_i = \mathrm{d_H}(\mathbf{r}_i, \hat{\mathbf{c}}^\star_i) = \mathrm{d_H}(\mathbf{r}_i, \hat{\mathbf{c}}_i) = d_i$ and $w_i = \mathrm{d_H}(\mathbf{r}_i, \mathbf{c}_i) \geq \mathrm{d_H}(\mathbf{r}_i, \hat{\mathbf{c}}'_i) = d'_i$. □

The key property of $\tilde{\alpha}_i$ is that it coincides with α_i upon correct decoding and may only *overestimate* the reliability upon decoding failure. As a consequence, any upper bound on $\Pr\big[\tilde{\alpha}_i \geq x, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big]$ is also a valid upper bound on $\Pr\big[\alpha_i \geq x, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big]$, and any lower bound on $\Pr\big[\tilde{\alpha}_i \geq x, \hat{\mathbf{c}}_i = \mathbf{c}_i\big]$ is a valid lower bound on $\Pr\big[\alpha_i \geq x, \hat{\mathbf{c}}_i = \mathbf{c}_i\big]$. Both directions are conservative for DFR estimation.

Theorem 1. *Denote as* $\bar{P}_{\text{in}}$ *an upper bound on the inner failure probability,* $\Pr\big[\hat{\mathbf{c}}_i \neq \mathbf{c}_i\big] \leq \bar{P}_{\text{in}}$*, such that* $\sum_{w_i} \Pr\big[|\mathbf{e}_i|_{\text{H}} = w_i\big]\tilde{\mathrm{P}}(\leq w_i - 1 \,|\, w_i) \leq \bar{P}_{\text{in}}$*, with*

$$\tilde{\mathrm{P}}(\leq w_i^\star \,|\, w_i) \triangleq \min\left\{ \sum_{\mathbf{c}\in\mathcal{B}\setminus\{\mathbf{0}\}} \Pr[\mathrm{d}_{\text{H}}(\mathbf{e}_i, \mathbf{c}_i) \leq w^\star \,|\, |\mathbf{e}_i|_{\text{H}} = w_i], 1 \right\}.$$

Then, for all $i \in [n_{\text{out}}]$ *and* $\tilde{\alpha}_i$ *as defined above, we have*

$$\begin{aligned}\Pr\big[\alpha_i \geq x, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big] &\leq \Pr\big[\tilde{\alpha}_i \geq x, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big] \leq \Pr\big[\alpha_i' \geq x, E_i'\big],\\ \Pr\big[\alpha_i \geq x, \hat{\mathbf{c}}_i = \mathbf{c}_i\big] &= \Pr\big[\tilde{\alpha}_i \geq x, \hat{\mathbf{c}}_i = \mathbf{c}_i\big] \geq \Pr\big[\alpha_i' \geq x, C_i'\big],\end{aligned}$$

with the definitions

$$\begin{aligned}\Pr\big[\alpha_i' \geq x, C_i'\big] &\triangleq \begin{cases} \sum_{w_i=0}^{n_{\text{in}}} \Pr\big[|\mathbf{e}_i|_{\text{H}} = w_i\big]\tilde{\mathrm{P}}(\leq w_i - x \,|\, w_i) & \textit{for } x > 0,\\ \bar{P}_{\text{in}} & \textit{otherwise,}\end{cases}\\ \Pr\big[\alpha_i' \geq x, E_i'\big] &\triangleq \begin{cases} \sum_{w_i=0}^{n_{\text{in}}} \Pr\big[|\mathbf{e}_i|_{\text{H}} = w_i\big]\left(1 - \tilde{\mathrm{P}}(< w_i + x \,|\, w_i)\right) & \textit{for } x > 0,\\ 1 - \bar{P}_{\text{in}} & \textit{otherwise.}\end{cases}\end{aligned}$$

Proof. As $\tilde{\alpha}_i$ is by definition non-negative,

$$\begin{aligned}\Pr\big[\tilde{\alpha}_i \geq 0, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big] &= \Pr\big[\hat{\mathbf{c}}_i \neq \mathbf{c}_i\big] \leq \Pr\big[\alpha_i' \geq 0, E_i'\big] = \bar{P}_{\text{in}}\\ \Pr\big[\tilde{\alpha}_i \geq 0, \hat{\mathbf{c}}_i = \mathbf{c}_i\big] &= \Pr\big[\hat{\mathbf{c}}_i = \mathbf{c}_i\big] \geq \Pr\big[\alpha_i' \geq 0, C_i'\big] = 1 - \bar{P}_{\text{in}}.\end{aligned}$$

For $\tilde{\alpha}_i \geq x > 0$, we show the case of $\hat{\mathbf{c}}_i \neq \mathbf{c}_i$, the case $\hat{\mathbf{c}}_i = \mathbf{c}_i$ is analogous. Denote the random variables $W_i = |\mathbf{e}_i|_{\text{H}}$ and $W_i^\star = \mathrm{d}_{\text{H}}(\hat{\mathbf{c}}_i^\star, \mathbf{e}_i)$. Since $\tilde{\alpha}_i = |W_i^\star - W_i|$, the event $\{\tilde{\alpha}_i \geq x,\, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\}$ decomposes as

$$\begin{aligned}\Pr\big[\tilde{\alpha}_i \geq x, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big] &= \Pr\big[(W_i^\star - W_i \geq x \cup W_i - W_i^\star \geq x), \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big]\\ &= \Pr\big[W_i - W_i^\star \geq x, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big],\end{aligned}$$

where the second equality holds because $W_i^\star - W_i \geq x > 0$ would imply $\hat{\mathbf{c}}_i = \mathbf{c}_i$ (the transmitted codeword is closer than all others). On the other hand, $W_i - W_i^\star \geq x > 0$ already implies $\hat{\mathbf{c}}_i \neq \mathbf{c}_i$, so the intersection is redundant and it follows that

$$\begin{aligned}\Pr\big[W_i - W_i^\star \geq x, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big] &= \Pr\big[W_i^\star \leq W_i - x\big]\\ &= \sum_{w_i} \Pr[W_i^\star \leq w_i - x \,|\, W_i = w_i] \Pr\big[W_i = w_i\big]\end{aligned}$$

The result follows using the union bound:

$$\Pr[W_i^\star \leq w_i - x \,|\, W_i = w_i] \leq \sum_{\mathbf{v}\in\mathcal{B}\setminus\{\mathbf{0}\}} \Pr[\mathrm{d}_{\text{H}}(\mathbf{v}, \mathbf{e}_i) \leq w_i - x \,|\, W_i = w_i].$$

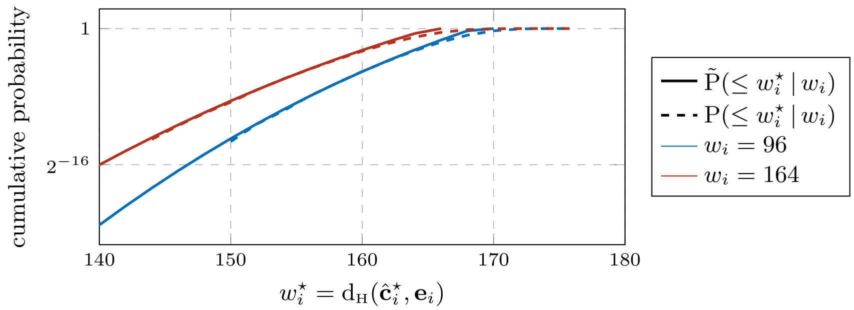

Fig. 1: Cumulative distribution of $w_i^\star$ for inner $\mathcal{B} = \mathrm{RM}$ with $n_{\mathrm{in}} = 384$; $p = 0.34$.

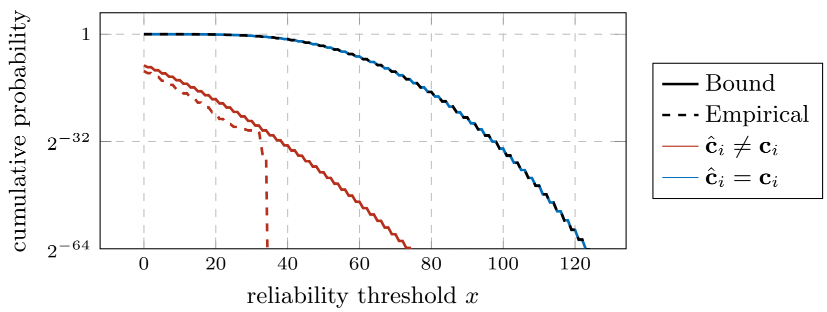

Fig. 2: Distribution of α_i for the inner code $\mathcal{B} = \mathrm{RM}$ with $n_{\mathrm{in}} = 384$ and $p = 0.34$.

The quantity $\tilde{\mathrm{P}}(\leq w^\star \mid w)$ is a simple union bound on the probability that at least one nonzero codeword lies within distance $w^\star$ of the error pattern and can be evaluated in closed form for arbitrary inner codes. The bound $\bar{P}_{\mathrm{in}}$ may also be derived via union bound arguments; the case of (duplicated) first-order RM codes is covered in [11].

Simulation results. Figure 1 compares the bound $\tilde{\mathrm{P}}(\leq w_i^\star \mid w_i)$ from Theorem 1 with the empirically estimated CDF $\mathrm{P}(\leq w_i^\star \mid w_i) \triangleq \Pr[\mathrm{d_H}(\hat{\mathbf{c}}_i^\star, \mathbf{e}_i) \leq w_i^\star \mid w_i]$ for fixed error weights $w_i \in \{96, 164\}$. For both values (as well as other tested error weights), the union bound $\tilde{\mathrm{P}}(\leq w_i^\star \mid w_i)$ provides a tight upper bound on the empirical distribution, shifting mass toward smaller values of $w_i^\star$ as expected.

Figure 2 compares the bounds on $\Pr\big[\tilde{\alpha}_i \geq x, \hat{\mathbf{c}}_i = \mathbf{c}_i\big]$ and $\Pr\big[\tilde{\alpha}_i \geq x, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big]$ from Theorem 1 with the empirical distributions of $\Pr\big[\alpha_i \geq x, \hat{\mathbf{c}}_i = \mathbf{c}_i\big]$ and $\Pr\big[\alpha_i \geq x, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\big]$. For the erroneous-decoding event, the analytical curve is a conservative upper bound on the empirical one for all x, confirming the validity of the bound from Theorem 1. For the correct-decoding event, the analytical and empirical curves are visually indistinguishable on the logarithmic scale; their difference is below 10^{-3} throughout. For α_i such that the Monte Carlo estimates are accurate to within 10^{-3}, the empirical values are consistently above the analytical lower bound, as expected.

4.2 Threshold-based Error-Erasure Decoding

This section analyzes the classical decoding strategy in which outer symbols whose inner decoding is deemed unreliable are *erased* before outer decoding [18]. This reduces the DFR by converting potential errors (which cost the outer code two units of minimum distance each) into erasures (which cost only one).

Erasure assignment scheme. Fix a threshold $T \in \mathbb{Z}$. After inner decoding, each outer symbol $i \in [n_{\mathrm{out}}]$ is classified as follows:

- If $\alpha_i \geq T$, the estimate $\hat{\mathbf{c}}_i$ is *accepted* and forwarded to the outer decoder.
- If $\alpha_i < T$, the position i is declared an *erasure*.

The outer decoder then operates in errors-and-erasures mode.

Error and erasure probabilities. Assuming $\mathbf{e} \sim \mathrm{Bern}(p)^n$, the inner blocks are independent and identically distributed. The decoding outcome at each position i is determined by three mutually exclusive events:

1. *Correct acceptance*, with probability $p_{\mathrm{c}}(T) \triangleq \Pr\left[\alpha_i \geq T,\, \hat{\mathbf{c}}_i = \mathbf{c}_i\right]$.
2. *Undetected error*, with probability $p_{\mathrm{e}}(T) \triangleq \Pr\left[\alpha_i \geq T,\, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\right]$.
3. *Erasure*, with probability $p_{\mathrm{s}}(T) \triangleq \Pr\left[\alpha_i < T\right] = 1 - p_{\mathrm{c}}(T) - p_{\mathrm{e}}(T)$.

Using Theorem 1, we bound the probabilities of Case 1 and 2 as

$$p_{\mathrm{e}}(T) = \Pr\left[\alpha_i \geq T,\, \hat{\mathbf{c}}_i \neq \mathbf{c}_i\right] \leq \Pr\left[\alpha_i' \geq T,\, E_i'\right] \triangleq p_{\mathrm{e}}'(T), \tag{3}$$

$$p_{\mathrm{c}}(T) = \Pr\left[\alpha_i \geq T,\, \hat{\mathbf{c}}_i = \mathbf{c}_i\right] = \Pr\left[\alpha_i' \geq T,\, C_i'\right] \triangleq p_{\mathrm{c}}'(T). \tag{4}$$

To cover Case 3, we further define

$$p_{\mathrm{s}}'(T) \triangleq 1 - p_{\mathrm{e}}'(T) - p_{\mathrm{c}}'(T). \tag{5}$$

In general, $p_{\mathrm{s}}'(T)$ is not guaranteed to be a lower or upper bound on $p_{\mathrm{s}}(T)$. However, for the DFR bound, $p_{\mathrm{c}}'(T) \leq p_{\mathrm{c}}(T)$ and $p_{\mathrm{e}}'(T) \geq p_{\mathrm{e}}(T)$ is sufficient.

Proposition 1. *For any threshold $T \geq 0$, the DFR is bounded as*

$$\mathrm{DFR}(T) \leq \mathrm{DFR}'(T) \triangleq \sum_{\substack{e,s \geq 0 \\ 2e+s \geq d_{\mathrm{out}}}} \binom{n_{\mathrm{out}}}{e,\, s} p_{\mathrm{e}}'(T)^e\, p_{\mathrm{s}}'(T)^s\, p_{\mathrm{c}}'(T)^{n_{\mathrm{out}}-e-s},$$

where the multinomial coefficient is $\binom{n_{\mathrm{out}}}{e,s} = \frac{n_{\mathrm{out}}!}{e!\, s!\, (n_{\mathrm{out}}-e-s)!}$ for $e + s \leq n_{\mathrm{out}}$.

Proof. Denote the number of undetected errors (Case 2) as e and the number of erasures (Case 3) as s, i.e., there are $n_{\mathrm{out}} - e - s$ correct acceptances (Case 1). The outer decoding fails if and only if $2e + s \geq d_{\mathrm{out}}$. Since each position corresponds to an independent multinomial trial with respect to the three possible outcomes, the DFR is computed as

$$\mathrm{DFR}(T) = \sum_{\substack{e,s \geq 0 \\ 2e+s \geq d_{\mathrm{out}}}} \binom{n_{\mathrm{out}}}{e,\, s} p_{\mathrm{e}}(T)^e \left(1 - p_{\mathrm{c}}(T) - p_{\mathrm{e}}(T)\right)^s p_{\mathrm{c}}(T)^{n_{\mathrm{out}}-e-s},$$

Replacing $p_\mathrm{e}(T)$ by $p'_\mathrm{e}(T)$ reduces the erasure probability in favor of the error probability, and replacing $p_\mathrm{c}(T)$ by $p'_\mathrm{c}(T)$ reduces the correct acceptance probability in favor of the erasure probability. The substitutions can therefore only increase the probability of incorrect outer decoding. □

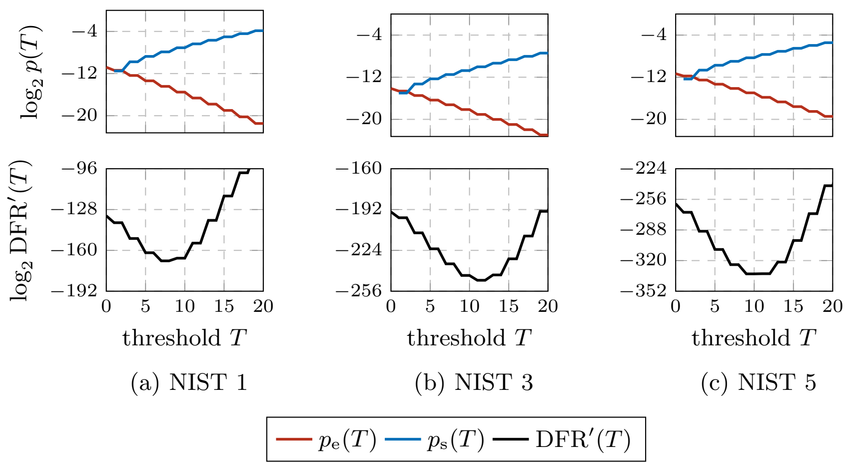

Fig. 3: Probability of inner error, inner erasure, and decryption failure as trade-off via threshold T. Concatenation of RM and RS code as in [11], see Table 2.

Choice of threshold. As illustrated in Figure 3 for code parameters suggested in [11], the threshold T controls the trade-off between errors and erasures. For $T = 0$, no erasures are declared, and the hard-decision decoder employed in HQC is recovered. As T increases, inner decoding failures with low reliability are increasingly caught as erasures, at the cost of increasing the erasure probability. We pick the threshold $T^\star$ that minimizes the DFR bound, i.e.,

$$T^\star = \arg\min_{T \geq 0} \ \mathrm{DFR}'(T).$$

Here, $T^\star = 7$ guarantees a DFR of at most 2^{-168}. Typically, a slightly higher or lower threshold offers a similar DFR performance.

Practical considerations. The threshold-based erasure assignment scheme proposed in this section (and also used in HARE [17]) inherently results in a varying number of erasures. This may be problematic in practice, since the number of erasures is secret-key dependent and may leak via side channels. The following section proposes an assignment scheme with a fixed number of erasures which achieves comparable performance (cf. Section 5).

4.3 Sorting-based Error-Erasure Decoding

An alternative to the threshold-based scheme of Section 4.2 is to erase a *fixed number* of outer symbols—specifically, those with the lowest reliability [18]. By fixing the number of erasures rather than the threshold, the erasure count becomes deterministic, which avoids worst-case fluctuations and simplifies the interaction with the outer code's error-correction budget.

Erasure assignment. Fix a parameter $\epsilon \in \{0, 1, \ldots, n_{\mathrm{out}}\}$. After inner decoding, partition $[n_{\mathrm{out}}] = \mathcal{J} \cup \mathcal{I}$ with $|\mathcal{I}| = \epsilon$, $|\mathcal{J}| = n_{\mathrm{out}} - \epsilon$, such that

$$\alpha_i \leq \alpha_j \quad \forall\, i \in \mathcal{I},\ j \in \mathcal{J},$$

with ties broken arbitrarily. The positions indexed by $\mathcal{I}$ are declared erasures, and the remaining $n_{\mathrm{out}} - \epsilon$ symbols are forwarded to the outer decoder in errors-and-erasures mode.

DFR via order statistics. The following DFR bound adapts the approach of Agrawal and Vardy [1] (see also [10]) to the HQC setting.

Let the outer code have minimum distance d_{out}. With ϵ erasures, the outer decoder can correct up to $t_\epsilon \triangleq \lfloor (d_{\mathrm{out}} - 1 - \epsilon)/2 \rfloor$ additional errors. Denote by $\bar{P}_{\mathrm{in}} \geq \Pr\left[\hat{\mathbf{c}}_i \neq \mathbf{c}_i\right]$ an upper bound on the inner decoding error probability. Then, the DFR bounded and decomposed as

$$\mathrm{DFR}(\epsilon) \leq \sum_{w=0}^{n_{\mathrm{out}}} \binom{n_{\mathrm{out}}}{w} \bar{P}_{\mathrm{in}}^{w} (1 - \bar{P}_{\mathrm{in}})^{n_{\mathrm{out}} - w} \cdot P_w(\epsilon), \tag{6}$$

where $P_w(\epsilon)$ denotes an upper bound on the outer failure probability conditioned on w erroneous inner decodings. Since the following lemma is directly adapted from [1], we omit a proof.

Lemma 2. *Let $\beta_1(w) \geq \cdots \geq \beta_w(w)$ denote the ordered reliabilities of the w erroneous positions and $\gamma_1(n_{\mathrm{out}} - w) \geq \cdots \geq \gamma_{n_{\mathrm{out}}-w}(n_{\mathrm{out}} - w)$ those of the $n_{\mathrm{out}} - w$ correct positions. Then, the outer failure probability is upper-bounded by*

$$P_w(\epsilon) = \begin{cases} 0, & \textit{if } w \leq t_\epsilon, \\ \Pr\left[\beta_{t_\epsilon+1}(w) \geq \gamma_{n_{\mathrm{out}}-\epsilon-t_\epsilon}(n_{\mathrm{out}} - w)\right], & \textit{if } t_\epsilon < w \leq \epsilon + t_\epsilon, \\ 1, & \textit{if } w > \epsilon + t_\epsilon. \end{cases} \tag{7}$$

Computing P_w via order-statistic CDFs. For any j, ℓ, the probability $\Pr\left[\beta_j \geq \gamma_\ell\right]$ can be expressed as

$$\Pr\left[\beta_j \geq \gamma_\ell\right] = \sum_{x=0}^{n_{\mathrm{in}}} \Pr\left[\gamma_\ell(n_{\mathrm{out}} - w) = x\right] \Pr\left[\beta_j(w) \geq x\right].$$

The required PMFs and CDFs of the order statistic can be computed from

$$\begin{aligned} \Pr\left[\beta_j(w) \geq x\right] &= \Phi_j^w(\Pr[\alpha_i \geq x \,|\, \hat{\mathbf{c}}_i \neq \mathbf{c}_i]), \\ \Pr\left[\gamma_\ell(n_{\mathrm{out}} - w) \geq x\right] &= \Phi_\ell^{n_{\mathrm{out}}-w}(\Pr[\alpha_i \geq x \,|\, \hat{\mathbf{c}}_i = \mathbf{c}_i]) \end{aligned}$$

where $\Phi_k^t(p) \triangleq \sum_{i=k}^{t} \binom{t}{i} p^i (1-p)^{t-i}$ denotes the binomial tail probability.

Note that the exact CDFs $\Pr[\alpha_i \geq x \,|\, \hat{\mathbf{c}}_i \neq \mathbf{c}_i]$ and $\Pr[\alpha_i \geq x \,|\, \hat{\mathbf{c}}_i = \mathbf{c}_i]$ are not available to use. The following proposition shows that we can rely on Theorem 1 to get a conservative bound on the DFR.

Proposition 2. *Let* $\bar{P}_{\text{in}} = \Pr\left[\alpha'_i \geq 0, E'_i\right]$ *be a bound on the inner failure probability and define*

$$F'_{\text{e}}(x) \triangleq \frac{\Pr\left[\alpha'_i \geq x, E'_i\right]}{\bar{P}_{\text{in}}}, \qquad F'_{\text{c}}(x) \triangleq \frac{\Pr\left[\alpha'_i \geq x, C'_i\right]}{1 - \bar{P}_{\text{in}}}$$

$$P'_w(\epsilon) \triangleq \sum_{x=0}^{n_{\text{in}}} \left(\Phi_\ell^{n_{\text{out}}-w}(F'_{\text{c}}(x)) - \Phi_\ell^{n_{\text{out}}-w}(F'_{\text{c}}(x+1))\right) \Phi_j^w(F'_{\text{e}}(x))$$

Then it holds that

$$\mathrm{DFR}(\epsilon) \leq \mathrm{DFR}'(\epsilon) = \sum_{w=0}^{n_{\text{out}}} \binom{n_{\text{out}}}{w} \bar{P}_{\text{in}}^w (1 - \bar{P}_{\text{in}})^{n_{\text{out}}-w} \cdot P'_w(\epsilon),$$

Proof. Using $\Pr\left[E'_i\right] = \bar{P}_{\text{in}}$, we get

$$\bar{P}_{\text{in}}^w \cdot \Phi_j^w(F'_{\text{e}}(x)) = \Phi_j^w(\bar{P}_{\text{in}} \cdot F'_{\text{e}}(x)),$$
$$\left(1 - \bar{P}_{\text{in}}\right)^{n_{\text{out}}-w} \cdot \Phi_l^{n_{\text{out}}-w}(F'_{\text{c}}(x)) = \Phi_l^{n_{\text{out}}-w}((1 - \bar{P}_{\text{in}}) \cdot F'_{\text{c}}(x)).$$

Now we can rewrite $\mathrm{DFR}'(\epsilon)$ as,

$$\sum_{w=t_\epsilon}^{\epsilon+t_\epsilon} \binom{n_{\text{out}}}{w} \sum_{x=0}^{n_{\text{in}}} \Phi_j^w(\bar{P}_{\text{in}} F'_{\text{e}}(x)) \left[\Phi_l^{n_{\text{out}}-w}((1 - \bar{P}_{\text{in}}) F'_{\text{c}}(x)) - \Phi_l^{n_{\text{out}}-w}((1 - \bar{P}_{\text{in}}) F'_{\text{c}}(x+1))\right] \tag{8}$$

For any two random variables A and B, if $\Pr\left[A \geq x\right] \geq \Pr\left[B \geq x\right]$, then we say that A stochastically dominates B in the first order. Then, given a non-increasing function $g : \mathbb{R} \to \mathbb{R}$, it holds that $\mathbb{E}\left[g(A)\right] \leq \mathbb{E}\left[g(B)\right]$. Notice that in Equation (8), the inner sum is an expectation over the random variable $B = (\alpha'_i, C'_i)$ which is stochastically dominated by $A = (\alpha_i, \hat{\mathbf{c}}_i = \mathbf{c}_i)$. Correspondingly, $g(x) = \Phi_j^w(\Pr\left[\alpha'_i \geq x, E'_i\right])$, which is non-increasing in x. Using this fact,

$$\mathrm{DFR}'(\epsilon) \geq \sum_{w=t_\epsilon}^{\epsilon+t_\epsilon} \binom{n_{\text{out}}}{w} \sum_{x=0}^{n_{\text{in}}} \Pr\left[\gamma_\ell(n_{\text{out}} - w) = x\right] \Phi_j^w(\Pr\left[\alpha'_i \geq x, E'_i\right]).$$

Finally, using the fact that $\Phi_j^w(\cdot)$ is non-decreasing, we get that $\mathrm{DFR}'(\epsilon) \geq \mathrm{DFR}(\epsilon)$ □

Choice of erasure number. Figure 4 illustrates how the parameter ϵ controls the trade-off between errors and erasures. For $\epsilon = 0$, no erasures are used, and the classical hard-decision decoder of HQC is recovered. As ϵ increases, more potentially erroneous symbols are erased, which reduces P_w for moderate ϵ. For larger ϵ the remaining error-correction radius $t_\epsilon = \lfloor (d_{\text{out}} - 1 - \epsilon)/2 \rfloor$ is too low. For ϵ such that $\epsilon \equiv d_{\text{out}} \pmod 2$, $\epsilon \pm 1$ can correct strictly more error patterns than $\epsilon = \varepsilon$; this leads to the ripple of the curve in Figure 4. When proposing parameters, we pick $\epsilon^\star$, that minimizes the DFR bound, i.e.,

$$\epsilon^\star = \arg\min_\epsilon \mathrm{DFR}'(\epsilon).$$

Note that other choices may offer a similar performance in terms of DFR with a lower implementation overhead.

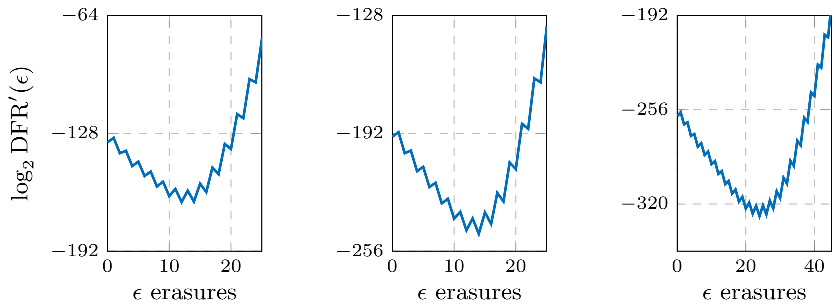

Fig. 4: DFR bound as a function of the number of erased positions ϵ. Concatenation of RM and RS code as in [11], see Table 2.

Table 2: Suggested parameters for NIST 1, 3, and 5. GCCs are constructed according to Table 1. Outer code and decoder parameters are chosen to guarantee a DFR of at most $2^{-\lambda}$ via Equation (2).

Security category	n	n_{in}	Outer RS				Outer $\mathcal{C}$ (BCH)			
			n_{RS}	k_{RS}	d_{RS}	$\mathsf{dec}(\cdot)$	$n_{\mathcal{C}}$	$k_{\mathcal{C}}$	$d_{\mathcal{C}}$	$\mathsf{dec}(\cdot)$
NIST 1	17 669	384	46	16	31	HDD	none [11]			
	16 901	384	44	15	30	$T^\star = 8$	44	25	8	$T^\star = 10$
	16 901	384	46	15	30	$\epsilon^\star = 13$	44	26	7	$\epsilon^\star = 4$
NIST 3	35 851	640	56	24	33	HDD	none [11]			
	34 589	640	54	24	31	$T^\star = 10$	54	30	9	$T^\star = 15$
	34 589	640	54	24	31	$\epsilon^\star = 12$	54	30	9	$\epsilon^\star = 4$
NIST 5	57 637	640	90	32	59	HDD	none [11]			
	55 541	640	86	31	56	$T^\star = 10$	86	43	14	$T^\star = 14$
	55 603	768	72	32	41	$\epsilon^\star = 16$	72	37	11	$\epsilon^\star = 7$

5 Parameters

For various NIST security levels, parameters relying on the proposed GCC (Section 3) and the reliability-based outer decoding (Section 4) are suggested in Table 2. Concretely, we provide results for both the threshold- and partition-based erasure assignment schemes. For a fixed RM and n_{out} value, we check the DFR of several RS codes of length n_{out}. Then, we perform a similar search for the second level to find a code $\mathcal{C}$ which meets the DFR requirement. By reducing n_{out}, we can reduce both the public-key length n bits (we choose the first primitive prime larger than $n_{\text{out}}n_{\text{in}}$) and the ciphertext length $n + n_{\text{out}}n_{\text{in}}$ bits. For NIST security level 1, we found that public-key and ciphertext lengths can be reduced by 4.34 %, which is on par with the improvements achieved in [4].

6 Conclusion

In this work, we proposed a two-level generalized concatenated code that exploits the repetition subcode within HQC's inner Reed–Muller code, together with a reliability-based errors-and-erasures decoding framework for the outer codes. We derived rigorous upper bounds on the DFR for both threshold-based and partition-based erasure assignment, and identified parameters that reduce public-key and ciphertext sizes by up to 4.34 % for NIST security level 1.

Several directions for further improvement are possible. The unbalanced encryption weights proposed in [17] are orthogonal to both our GCC structure and our decoder; combining them with the techniques of this work would yield additional size reductions. Similarly, ciphertext compression [4] is compatible with our construction and could further reduce bandwidth. Finally, our DFR analysis assumes i.i.d. Bernoulli errors; incorporating a more accurate error model for HQC [4, 6] may tighten the bounds and enable shorter codes.

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